

Final Project | Topics in Computer Science II | Fall 2025, University of Iowa

Comparison of Thread, Zigbee and Z-Wave for Home Automation

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Quick Note

This “presentation” contains many short form paragraphs in contrast to a typical bulleted format to more clearly explain ideas and thinking.

I felt this to be sufficient as I did not verbally present this material.

Agenda

1. Project Motivation
2. Protocols
 - a) Thread
 - b) Zigbee
 - c) Z-Wave
3. Comparison
4. Future Work (an automation I will build)
5. Conclusion
6. Thank You!

Project Motivation

What am I trying to achieve

Motivation pt1

After taking *Embedded Systems* course at the University of Iowa, I was opened to the possibilities of using microcontrollers to **develop smart home automations for my future apartment.**

Last summer, I collected an assortment of different devices with the intention of experimenting with wireless area networks and simple automations. Unfortunately, work was busy and **I never got around to accomplish anything meaningful with these devices.**



Motivation pt2

Fortunately, *Topics in Computer Science II: Real-Time Cyber Physical Systems* was offered during the fall semester, which immediately caught my eye.

After **learning the theory behind wireless network systems** from 802.11, 802.15.4, and 5G, participating in literary analysis through presentations, and reading multiple academic papers we were assigned a final project. The project goals are either a study or an implementation.

Having already purchased many of the necessary devices, I thought that there was **no better time study smart home protocols** to serve as **preliminary research** for my future personal smart home.

Project Goals

My goals of this study were simple:

- *Compare OSI models of protocols*
- *Compare protocol capabilities*
- *Highlight major considerations for each protocol*
- *Devise a simple automation that uses a suitable of choice*
- *Create a slideshow to aid in my future smart home development*

This project was intended to be a high-level comparison, drawing from many different sources – both academic and non-academic – to get a holistic view of chosen protocols.

Protocols Chosen

This project looks at the *architecture, usage, and personal* considerations of the following WLAN protocols:

- **Thread**
- **Zigbee**
- **Z-Wave**

Protocols that were not selected:

- WiFi: I require a dedicated, local network
- Bluetooth: limited by coverage range
- Wired: far too cumbersome and obtrusive

Summary of Protocols



Thread

- Thread Alliance formed 2014
- Mesh network
- Supports ~16000 devices
- Open standard
- Low power smart home devices



Zigbee

- Introduced 2005
- Mesh network
- Supports ~65000 devices
- Open standard
- Home automation, medical device data collection



Z-Wave

- Introduced 1999
- Mesh network
- Supports 232 devices
- Proprietary (Silicon Labs)
- Residential and commercial building automation

Thread

Protocol 1/3



Thread Background

- **Thread Group:** July 15, 2014 (Founded by Silicon Labs + six others)
- **OpenThread:** open source implementation (Founded by Google)
 - Protocol used for [Google Nest](#)

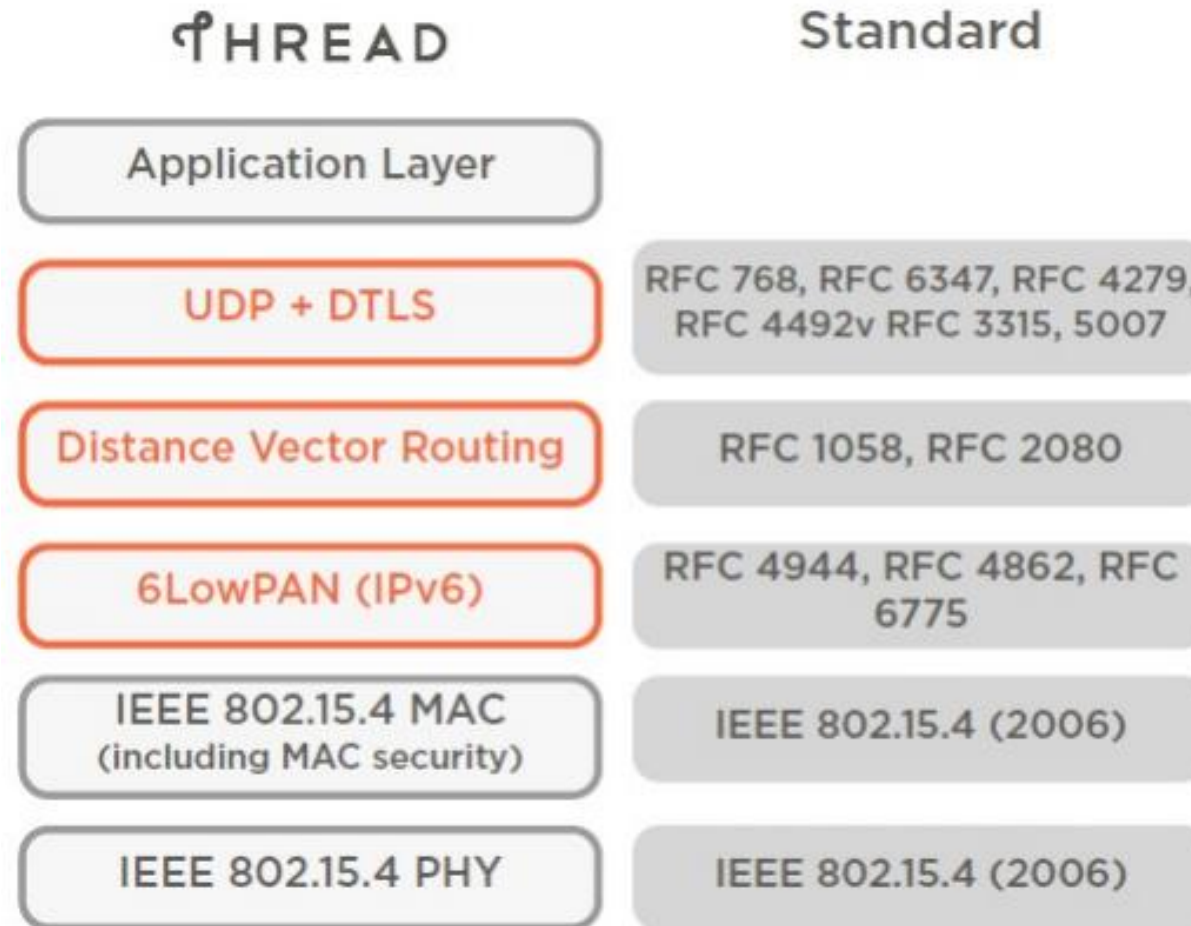
Features:

- IPv6 Over Low Power Wireless Personal Networks (6LoWPAN)
- Several to 100s of devices, scalable to 10,000s of devices
- Life Expectancy: years under typical conditions (powered by AA batteries or main power)
- Network can additionally function using WiFi or Ethernet

Thread Device Nomenclature

1. **Border Routers:** connect 802.15.4 to external WiFi/Ethernet networks
2. **Leader:** manages Thread network, assigns router IDs, and selects routers
3. **Thread Routers:** provides routing, joining, and security to network devices; cannot sleep
4. **Router-Eligible End Devices (REED):** can become Thread Router or Leader, however they cannot route messages
5. **End Devices:** not router-eligible; can be full end or minimal end devices
6. **Sleepy End Devices:** can communicate only through their Thread Router parent; cannot relay messages to other devices
7. **Synchronized Sleep End Devices:** Sleepy End Devices that use *CSL* for synchronized, low-power communication

Thread Architecture – Diagram



Thread Architecture – High Level

IEEE 802.15.4 PHYS:

- 2.4 GHz
- 250 kbps

IEEE 802.15.4 MAC:

- Carrier Sense Multiple Access (CSMA)
- MAC layer encryption enabled by higher layers

Thread's Optimizations:

- *Enhanced Frame Pending* – Acknowledges data with 'pending' status to save sleepy device battery
- *Enhanced Keepalive* – Data messages function as 'keepalives' reducing link maintenance traffic
- *Coordinated Sampled Listening (CSL)* – Synchronized transmit/receive periods for improved synchronicity and reduced power
- *Enhanced ACK Probing* – Reuses data for link metrics to provide granular control and to save energy

Thread Architecture – What Makes Thread Unique?

Thread is unique in that it provides IPv6 addressing!

Native IPv6 Support:

- End-to-end IPv6 addressing with no application layer translation
- Devices can communicate directly via IP
- Interoperability with IP-based ecosystems like WiFi and Ethernet

6LoWPAN Link Layer Forwarding:

- Header compression and fragmentation optimized for lowpower/constrained devices
- Efficient multi-hop forwarding

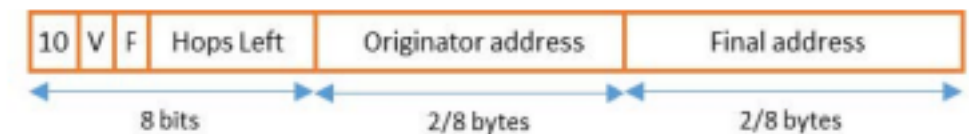


Figure 3.1. Mesh Header Format

Thread Routing

1. Distance-Vector Routing:

- **Next-Hop Selection:** each router maintains a table containing the best “next-hop” to another router
- **Hop Count Metric:** “distance” is measured as the number of hops between devices. Maximum is typically set by manager to ensure network efficiency
- **Periodic Updates:** routers regularly update routing costs for direct neighbors

2. Mesh Link Establishment (MLE):

- **Link Quality Sensing:** measures RSSI of incoming messages to assign “quality” [0-3]; 0 unknown cost
- **Bidirectional Verification:** ensures link is reliable in BOTH directions before including in routing table
- **Route Propagation:** Leader sends advertisement messages containing routing costs disseminated every few seconds

3. Retries (**note Thread uses UDP**)

- Employs MAC level retries and Application level retries

Currently, Thread supports <32 routers

Thread Joining & Network Operation

ALL JOINING IS USER INITIATED OVER WiFi

A device joins a Thread network using credentials via user mobile device. This authenticates securely and ensures device receives network parameters and security material.

Again, this is not something which I would like in my network as there is security threat for some remote access/authentication of devices which I did not authorize.

Once a device has joined, it uses MLE exchange to configure or exchange information about neighbors, device capabilities and device type, link costs, and security information.

After the joining and configuration process is complete, the mesh network is built.

Thread Usage – Home Mesh Network

External Connectivity:

- Border Routers bridge the Thread network to Wi-Fi or Ethernet

Network Management:

- A Leader manages router IDs and assigns network addresses

Message Routing:

- Thread Routers provide security services and relay data packets

Dynamic Roles:

- REEDs can automatically become routers if the network needs expansion

Device Communication:

- End devices communicate through a parent router to save power

Power Efficiency:

- Sleepy devices use synchronized schedules to minimize radio usage

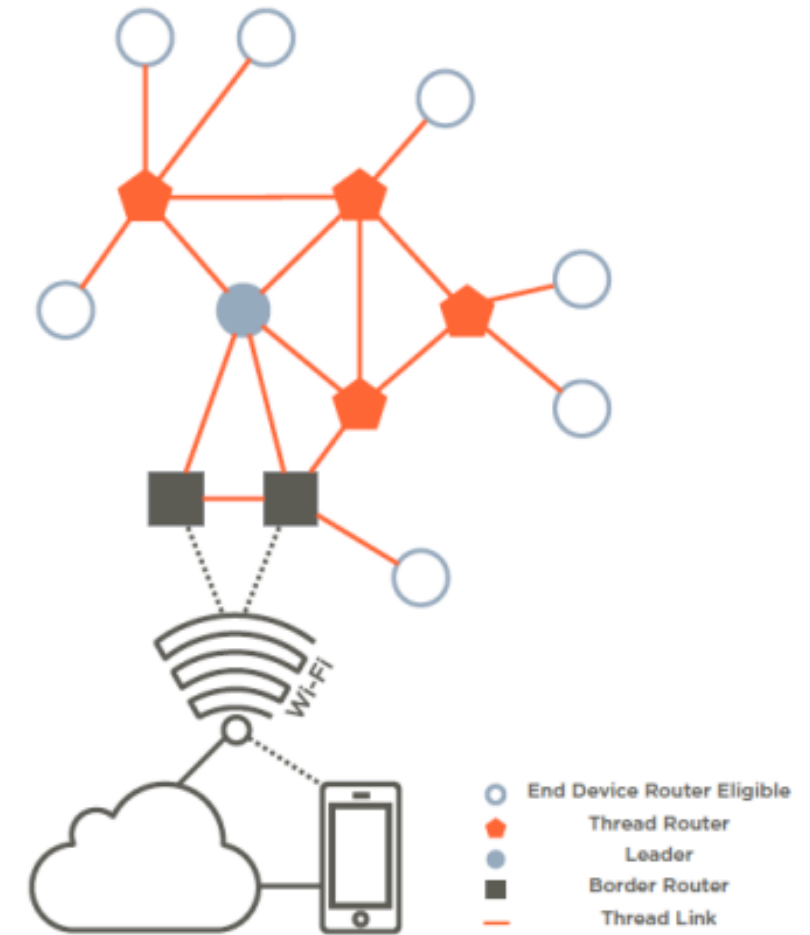


Figure 2.1. Thread Network Architecture

Thread Usage – Home Mesh Network

External Connectivity:

- Border Routers bridge the Thread network to Wi-Fi or Ethernet

Network Management:

- A Leader manages the network

Message Routing

- Thread Routers forward packets

Dynamic Roles:

- REEDs can automatically expand the network to meet needs

Device Communication:

- End devices communicate through a parent router to save power

Power Efficiency:

- Sleepy devices use synchronized schedules to minimize radio usage

User can control the Thread Network using a mobile device over internet!

This is concerning to me!

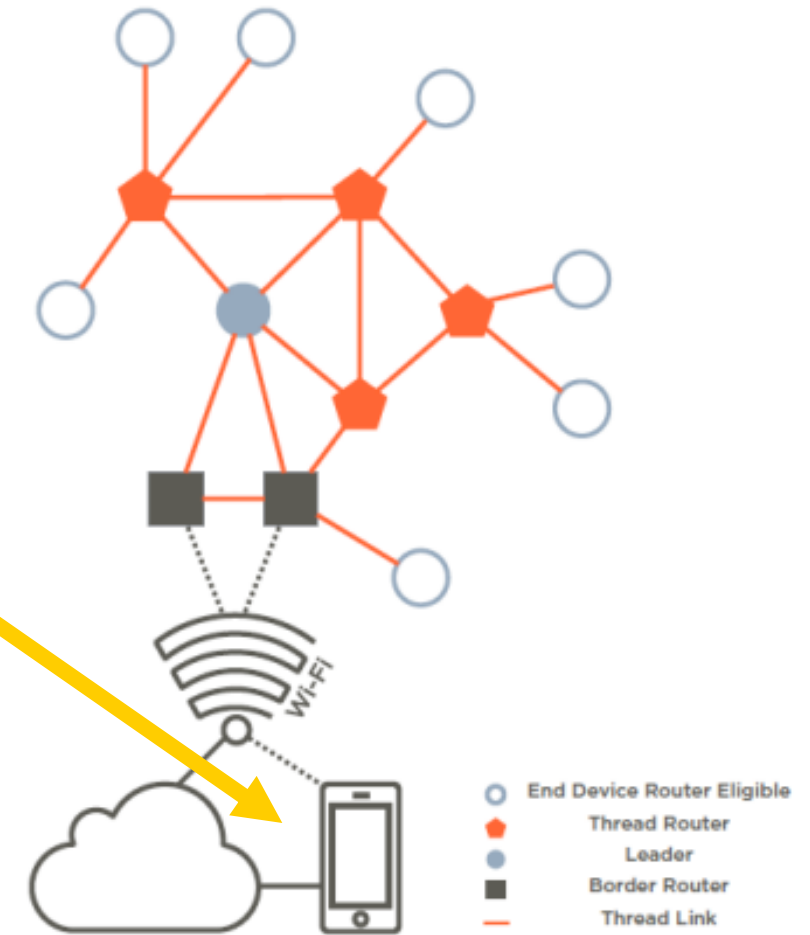


Figure 2.1. Thread Network Architecture

Thread – My Concerns + Security Guarantee?

As the **Thread Border Router** is accessed over WiFi, there is the inherent ability for an attacker to access my home network and thus all of my smart Thread devices. (not that it would be that catastrophic, but I would like control over my smart devices, etc 😊)

The Border Router is the main threat, as IPv6 != Open Access. Additionally, Thread protocol provides the following security assurances:

1. Secure against Over-the-Air (OTA) wireless attacks outside of the network, protected by 802.15.4-2006. Thread enhances this by exchanging frame counters in MLE handshaking.
2. Internet-Based Attacks: the network manager is protected by Datagram Transport Layer Security (DTLS) connection to the Border Router. Protected by passphrase

Yet, if my home network is compromised, I would hope my entire house isn't compromised too... This is a dealbreaker for me; I want my devices and automations to be controlled locally by me only.

Thread Performance

Latency (shown in figure): <200ms (this is super fast, “instant-on” human interaction)

Reliability: stable even in high-traffic multicast environments

Scalability: performance unwavering even as network grows

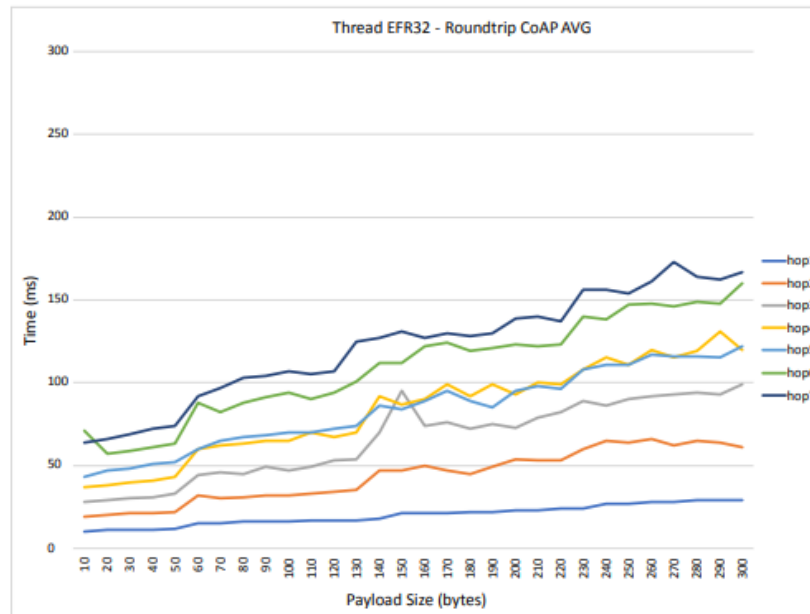


Figure 4.1. Thread EFR32 – Roundtrip CoAp AVG

This is the summary of extensive testing done by Silicon Labs: [Thread Mesh Network Performance](#)

Thread Membership

This is not a concern for me as I would be simply developing a smart home using Thread devices, however the [Thread Group](#) provides a six-tier membership.

The tiers distinguish organizations by how deeply they participate in Thread development and how much influence they have over specification, certification processes, and standard.

MEMBERSHIP BENEFITS BY TIER						
Member Benefits	Academic	Affiliate*	Associate**	Implementer	Contributor	Sponsor
Access To Members-Only Website	✓	✓	✓	✓	✓	✓
Use Of Alliance Member Logo	✓	✓	✓	✓	✓	✓
Participate In All-Member Sessions of General Meetings	✓	✓	✓	✓	✓	✓
Participate In Promotional Materials	✓	✓		✓	✓	✓
Access To Pre-Publication Draft Specification	✓	✓		✓	✓	✓
Access to IP Rights As Defined with Certification			✓	✓	✓	✓
Ability To Certify Devices By Inheritance For Approved Scenarios			✓	✓	✓	✓
Participate In Committee Sessions Of General Meetings				✓	✓	✓
Access To Thread Test Harness				\$5k/seat/yr	FREE	FREE
Ability To Pre-test And Certify Devices At Thread Group ATLs					✓	✓
Ability To Purchase Thread Test Bed					✓	✓
Access All In-Process Draft Specifications					✓	✓
Access To Thread Developed Apps					✓	✓
Authorize And Refer Associate Members					✓	✓
Participate And Vote In Work Groups and Committees					✓	✓
Chair Work Groups And Committees					✓	✓
Initiate Work Groups Or Committees						✓
Approve Operating Budget						✓
Approve Final Deliverables						✓
Automatic Seat On Board Of Directors						✓
MEMBERSHIP COSTS						
	Academic	Affiliate*	Associate**	Implementer	Contributor	Sponsor
Annual Fee	\$-	\$1,000	\$-	\$7,500	\$20,000	\$85,000
One-Time Initiation Fee						\$45,000

Thread Summary

Architecture:

- 802.15.4 + IPv6

Security:

- Thread Border Router accessible via internet

Performance:

- Extremely fast and scalable

Personal Comment:

- At this point in time, I do not want WiFi enabled smart devices. **Thread is not for me**



Zigbee

Protocol 2/3



Zigbee Background

- **Introduced:** June 13, 2005 (Founded by Silicon Labs + six others)
- **Connectivity Standards Alliance:** maintain and publish Zigbee and Matter standard (formed 2002, formerly Zigbee Alliance)

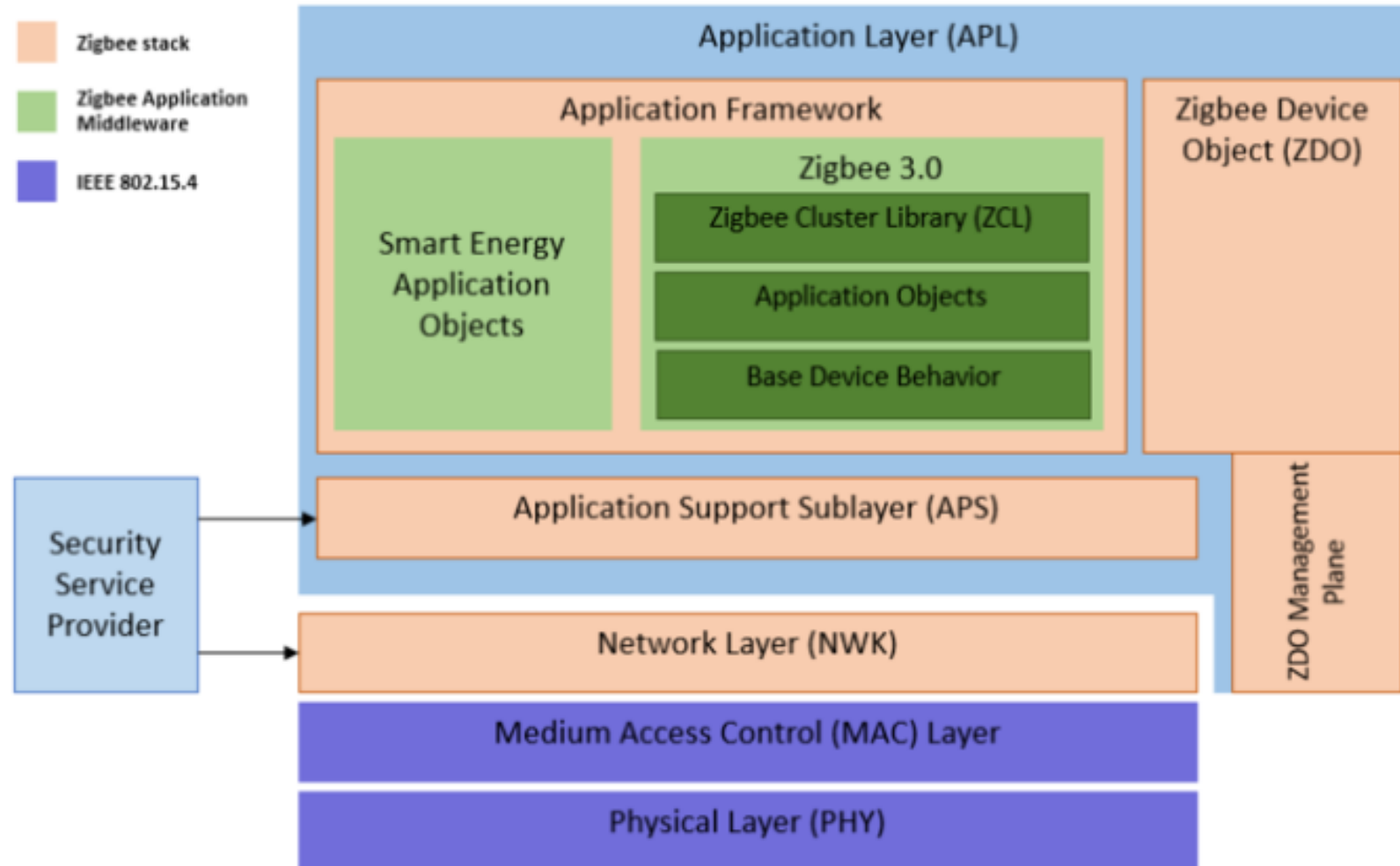
Features:

- Chips are integrated with radios and microcontrollers
- Includes network and application layers
- Zigbee Device Object (ZDO) provided to manage devices, keys, and policies
- Source routing

Zigbee Device Nomenclature

1. **Coordinator:** configures Zigbee network. Assigns and manages network addresses. Always-on “brain”.
2. **Router (relay):** full-function node that forwards packets to extend coverage and redundancy.
3. **End Device (leaf):** simple node that only sends and receives data via its parent (coordinator or router). Never forwards traffic for others.
4. **Gateway/Hub:** border device that combines Zigbee coordinator with IP interface to expose Zigbee mesh to external systems

Zigbee Architecture - Diagram



Zigbee Architecture – High Level

IEEE 802.15.4 PHYS:

- 2.4 GHz
- 250 kbps

IEEE 802.15.4 MAC:

- Carrier Sense Multiple Access (CSMA)

Zigbee's Optimizations:

Network Layer Mesh & Routing – adds multi-hop networking, address assignment, and self-healing routes

Application Support Sublayer – provides binding, addressing, mapping, and reliable delivery services. Acts as service bus for all applications and ZDO.

Zigbee Device Object (ZDO) – handles device and service discovery and role assignment. Central point for network management and security control.

Zigbee Architecture – What Makes Zigbee Unique?

Zigbee provides a comprehensive application layer! (It is also very mature)

Zigbee utilizes the Zigbee Cluster Library (ZCL)

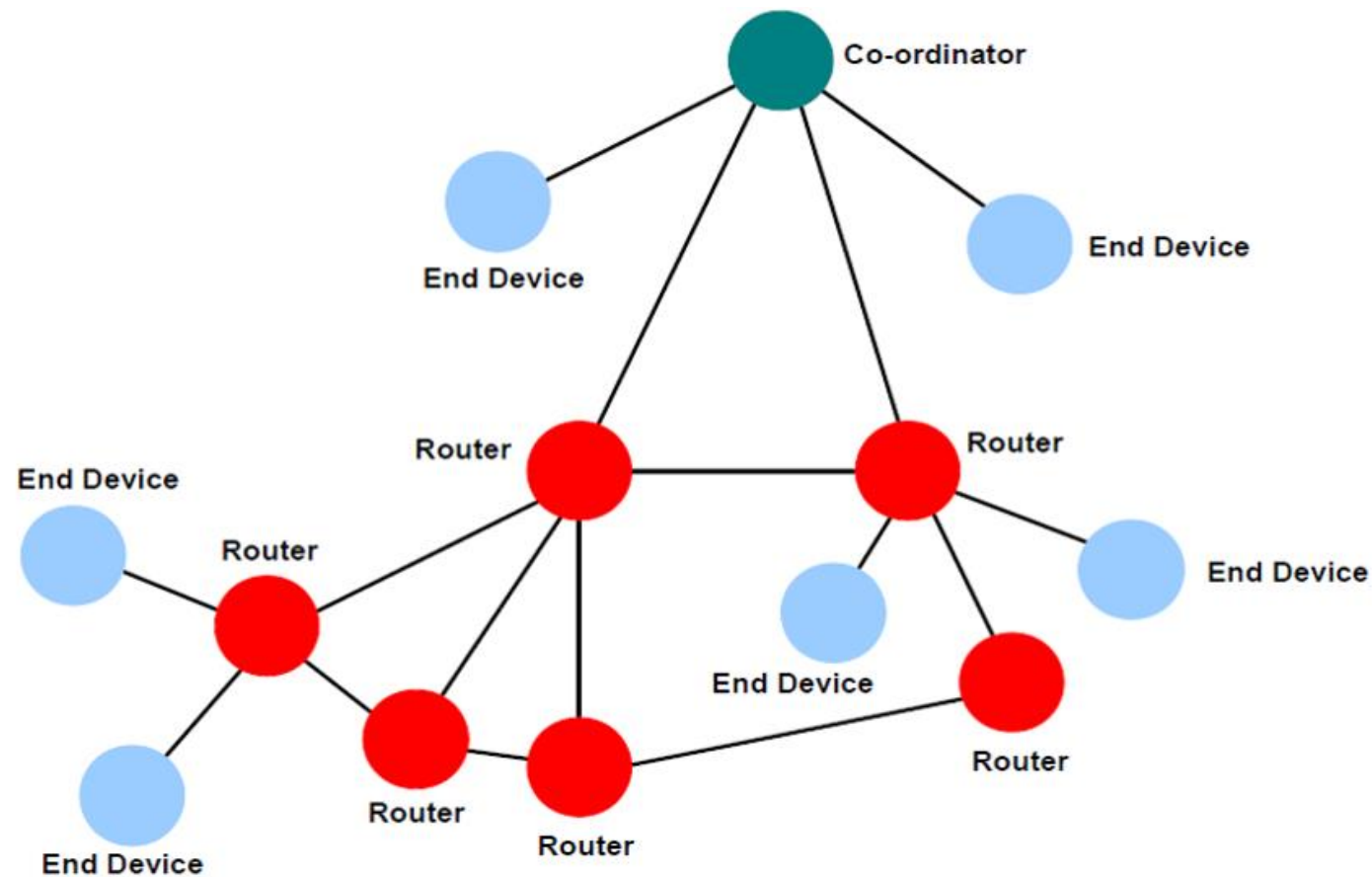
- *Built-in logic* – define behaviors for lights, sensors, and other devices
- *Direct Device Binding* – allows switches to control lights without a hub
- *Out-of-the-Box-Interoperability* – devices from different brands understand each other natively under ZCL

Ultimately, this **reduces the software complexity** as developers do not need to define custom formats or schemas.

Zigbee Routing

- **Reactive AODV Protocol:** ad-hoc on demand routing to find paths
- **Route Discovery (Broadcast):** source broadcasts requests to all neighbors to find destinations
- **Route Reply (Unicast):** nodes stores the next hop address and path costs
- **Source Routing:** packets carry full list of relays to destination

Zigbee Usage – Home Mesh Network



Zigbee Security

Strong AES-128 Cryptography:

- Zigbee secures data using symmetric AES-128 encryption in CCM* mode to provide confidentiality, integrity, authentication, and replay protection in network

Centralized Trust Center (TC):

- A trust center manages device authentication, key distribution, and access control

Key Hierarchy and Scope:

- Security relies on network keys (broadcast), link keys (unicast), and optional master keys for key establishment; improper key handling directly weakens network security

**** Key Management is the Main Weakness ****

- Compromise of a shared Network Key **can expose the entire network** 😞

Zigbee Performance

Zigbee shows similar performance to Thread

Latency (shown in figure): <200ms (this is super fast, “instant-on” human interaction)

Reliability: latency increases as packet payload increases (in contrast to Thread)

Scalability: performant up to 192 nodes, which is plenty for home networks

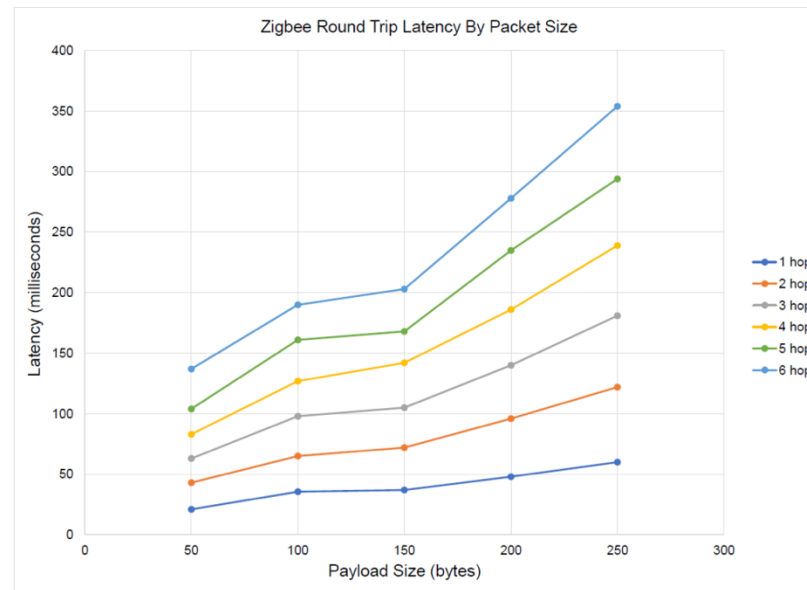


Figure 4.1. Zigbee Round Trip Latency by Packet Size

This is the summary of extensive testing done by Silicon Labs: [Zigbee Mesh Network Performance](#)

Zigbee Membership

This is not a concern for me as I would be simply developing a smart home using Zigbee devices, however the [Connectivity Standards Alliance](#) provides a four-tier membership.

Promoter	Participant	Adopter	Associate
*\$105,000 USD/yr	*\$20,000 USD/yr	\$7,000 USD/yr	\$0 USD/yr
Develop, Test, and Certify >	Develop, Test, and Certify >	Develop, Test, and Certify >	White label or rebrand a Certified Product via the Certification Transfer Program and use the Alliance Certification trademarks \$2,500 USD per product + \$500 USD per year, per product (due annually on the anniversary date of the grant of certification)
Go to Market >	Go to Market >	Go to Market >	
Participate and Lead >	Participate and Lead >	Participate and Lead >	

The tiers distinguish members by how much influence they have over the Matter (another protocol) and Zigbee standards and how deeply they can participate.

Zigbee Summary

Architecture:

- 802.15.4 + application layer

Security:

- Strong cryptography, but network key can compromise entire network

Performance:

- As fast as Thread, but not as scalable as Thread

Personal Comment:

- Zigbee satisfies my needs. Additionally, the provided application layer is an **added benefit**

Z-Wave

Protocol 3/3



Z-Wave Background

- **Introduced:** 1999 (developed by Zensys, acquired by Silicon Labs in 2018)
- **Z-Wave Alliance:** maintains and certifies the Z-Wave standard

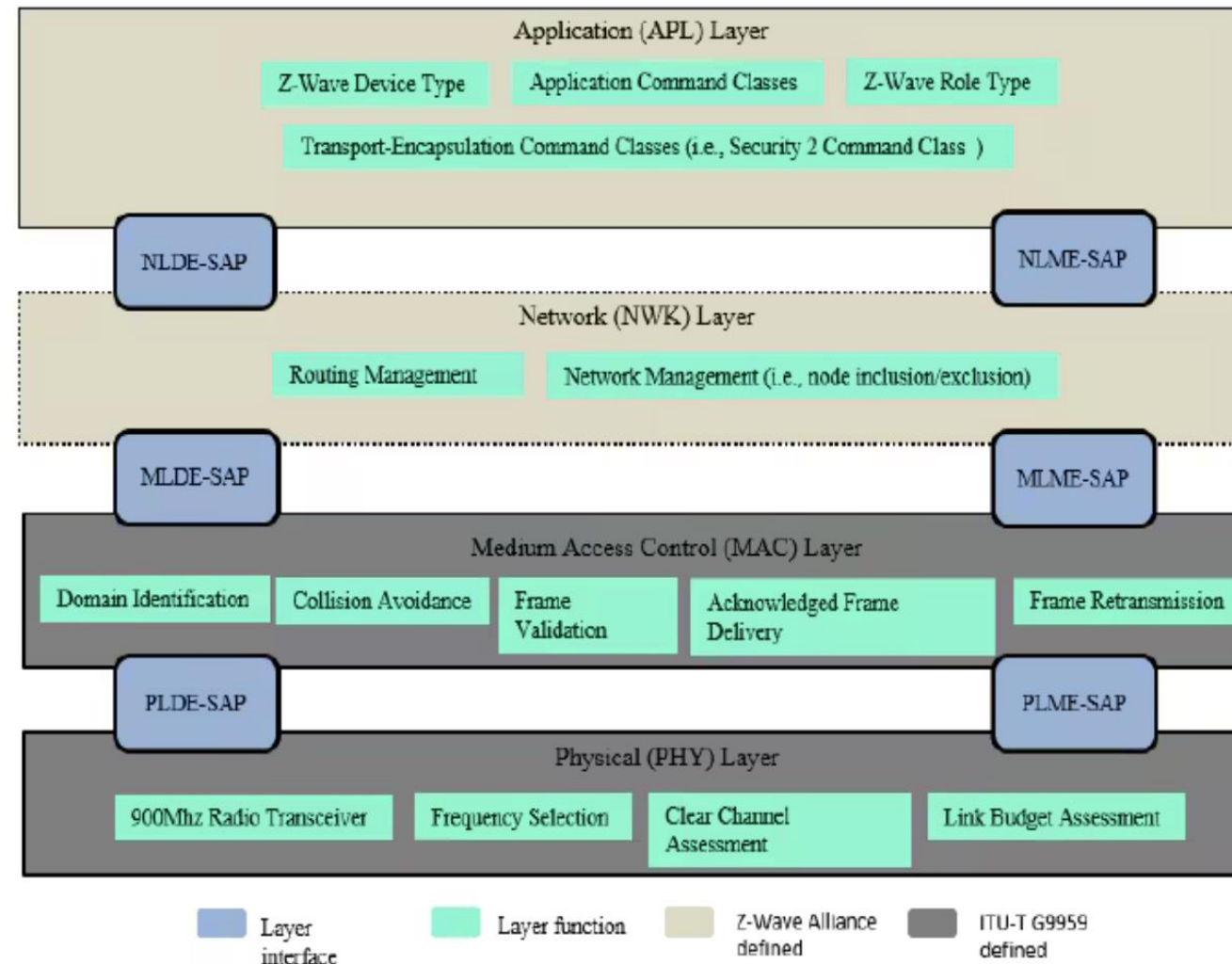
Features:

- Integrated RF transceiver with embedded protocol stack
- Includes MAC, networking, application layers
- Can be long range (up to 1.5mi)

Z-Wave Device Nomenclature

1. **Primary Controller:** configures and manages the Z-Wave network. There is only one primary controller that is always on. Assigns home ID and node IDs and maintains authoritative routing table.
2. **Slave Nodes:** execute commands and can additionally route traffic if always powered. However, battery powered slaves typically do not route.
3. **Gateway/Hub (VERA):** used to create scenes, events, timers. Used for remote control.

Z-Wave Architecture - Diagram



Z-Wave Architecture – High Level

Proprietary Z-Wave PHY/RF:

- Sub-GHz ISM band RF
- Optimized for reliable transmission through walls and floors

Proprietary Z-Wave MAC:

- Collision avoidance
- Data encoded using Manchester or PE coding

Z-Wave Optimizations:

- *Mesh & Routing* – provides multi-hop mesh networking using routing tables, automatic path discovery, and route repair (although maximum of 4 hops using source routing)
- *Transfer Layer* – ability for acks, retransmissions, checksum validation. Supports singlecast, multicast, and broadcast
- *Application Layer* – executes Z-Wave command classes, decodes payloads, and perform device specific actions (e.g. light switch or lock)

Z-Wave Architecture – What Makes Z-Wave Unique?

In particular, there are two reasons that makes Z-Wave unique:

1. Long-Range Z-Wave (Z-Wave LR):

Extends communication range well beyond classic Z-Wave mesh

Larger addressing space to support up to 4000 nodes

Very practical for automation within apartment complexes and hospitality markets

2. Z-Wave is a Close Source/Proprietary Protocol:

- The standard is governed by the Z-Wave alliance
- Strict certification and compliance testing
- A consequence of this is that it is also harder to understand/study – personal experience from this project

Because I understand this protocol less (proprietary vs based on 802.15.4), this protocol is not for me 😞

Z-Wave Routing

- **Mesh Routing Protocol:** uses multi-hop mesh routing to deliver frames beyond direct radio range
- **Controller-Based Path Discovery:** the primary controller maintains network topology and routing table built during analysis rather than on-demand
- **ACK-Based:** frames are forwarded hop-by-hop using routed singlecast with acknowledgments
- **Source Routing:** packets carry routing information defining the relay nodes to be used to reach the packet's destination. Maximum of 4 hops (although sources disagree on 4 or 5. Silicon Labs says 4).

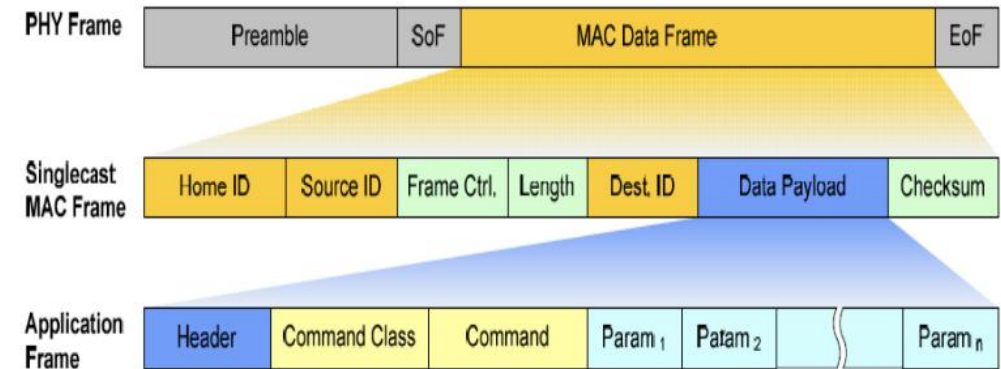
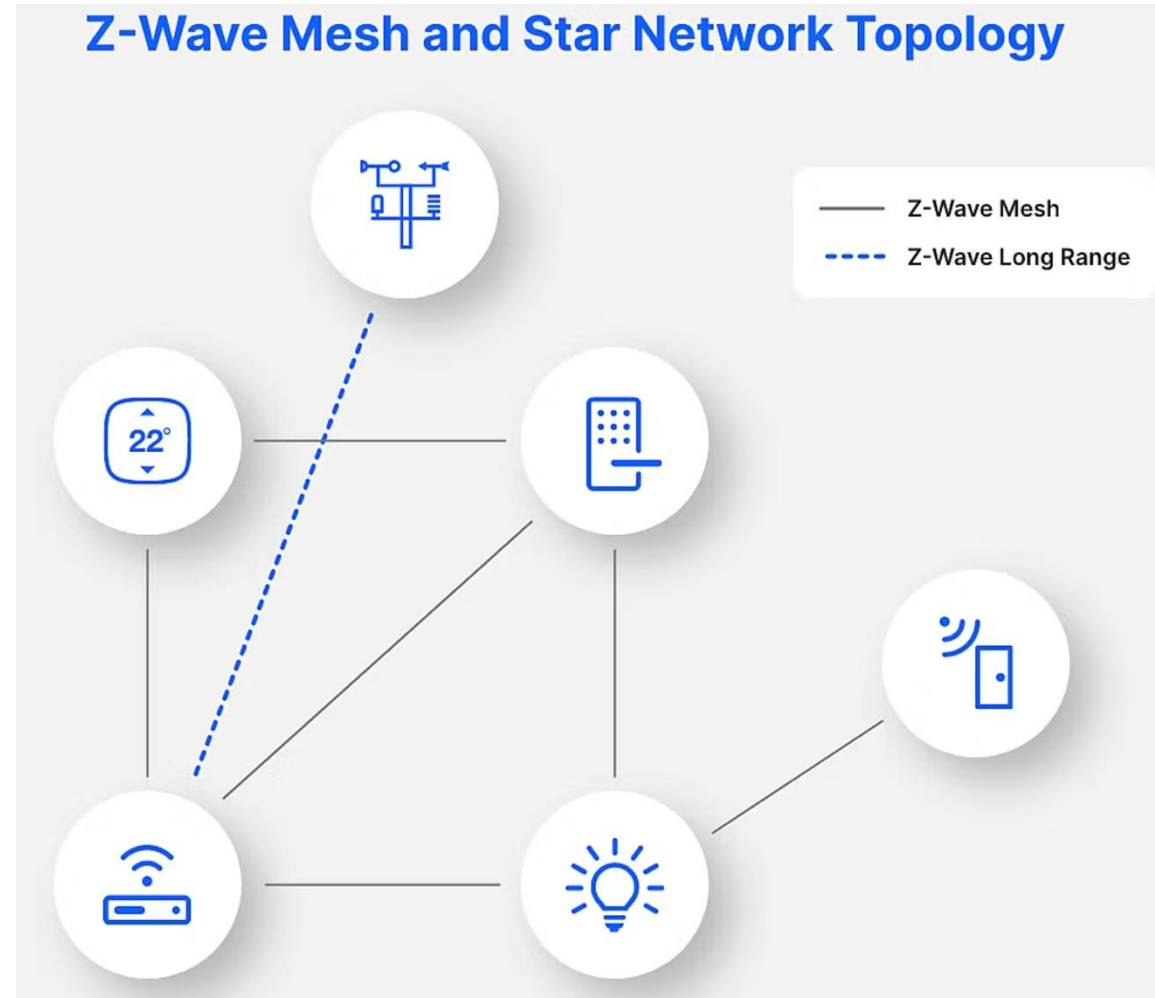


Figure 1. Z-Wave data in the different layers [9].

Z-Wave Usage – Home Mesh Network



Z-Wave Security

Application Layer Security: there are two security command classes: Security 0 (S0) and Security 2 (S2). S0 uses single shared network key while S2 introduces an authenticated key exchange

Secure Key Exchange: during secure inclusion, devices perform security bootstrapping using PIN code, QR, or smart-start methods

Access-Level Protection: S2 defines different security levels to match risk profiles

Z-Wave Performance

I could not find any paper/source that looks at the Z-Wave performance. Doing a quick glance, sentiments regarding Z-Wave performance are not positive:

Searching Z-Wave Performance:



Reddit · r/homeassistant

20+ comments · 3 years ago

Im kind of fed up with my poorly performing zwave network. ...



Hubitat

<https://community.hubitat.com> › is-zwave-slow-or-is-it-...

Is Zwave slow or is it just me? - 🔔 Get Help



Home Assistant Community

<https://community.home-assistant.io> › ... › Z-Wave

Massive unexplained Z-Wave performance drop in the last ...



samsung.com

<https://community.smarthings.com> › slow-zwave-and-g...

Slow Zwave, and getting slower - Devices & Integrations

Obviously, this is not a very good way to quantify performance...

Z-Wave Summary

Architecture:

- Proprietary Architecture

Security:

- S0 and S2 Security with multiple layers of security

Performance:

- I could not find any data/studies on the performance of Z-Wave

Personal Comment:

- Z-Wave was the most difficult protocol to research during this project. I believe this is because it is a proprietary protocol.

Comparison

High-level comparison and my personal selection

High Level Comparison – Decision Matrix

	Thread	Zigbee	Z-Wave
Maximum Devices	16384	65535	232
Devices on Market	300> (unverified)	4900>	4500>
Protocol Age	11 years	20 years	26 years
Development Kit	OpenThread	Expressif Zigbee	OpenZWave
Build Custom Device?	Yes	Yes	Yes
Open Standard	Yes	Yes	No
Protocols	802.15.4 + IP	802.15.4	Proprietary SoC
Data Rate	250 Kbps	250 Kbps	100 Kbps
Range	30 m	10-100 m	100 m – 1 mi
Frequency	2.4 GHz	2.4 GHz	868 MHz – 908 MHz
Fully Local Network	No	Yes	Yes

My Takeaways

The three protocols are all viable and proven options for smart home automation:

- **Zigbee** is the most proven smart home automation protocol, providing strong mesh network capabilities, large number of devices, and easy configuration.
- **Thread** is similar to Zigbee, but provides additional benefits of IP-addressability and multiple hosts. Simple sentiment analysis of YouTube videos is that Thread may overtake Zigbee.
- **Z-Wave** is another viable option that can enable longer ranges, however its closed standard and the Z-Wave Alliance requirement makes learning, understanding, and building Z-Wave automations more difficult.

For my personal needs, Zigbee is the clear winner:

- ✓ Ease of development and access to many devices
- ✓ Ability to create custom devices using ESP32
- ✓ Previous knowledge of 802.15.4 from Topics in Computer Science II
- ✓ Well known and easy to use application layer managers

Future Work

How I use my learning from this course & project

Smart Home Automation

This project has given me a theoretical understanding of smart home mesh networks protocols, and now **I would like to implement what I have learned.**

After graduation this may, I will be moving into a new home/apartment. This is the perfect opportunity for me to experiment with simple automations.

Smart Home Automation Goals

Ideally, I would like the ability to do the following:

- **Automate my mornings/nights:** lights, music, body heat regulation
- **Enhance home security:** camera, network anomaly detection
- **Everything runs locally:** limit outside access to the personal data

These are just a few... Too many ideas and not enough time! 😊

Future Work Roadmap



Smart Home
Study



This Project



Smart Home
Planning



Smart Home
Implementation

Example Plan

After conducting thorough research for this project, I will use **Zigbee** to implement my smart home due to its ubiquity and interoperability.

There are nearly **5000 supported devices from over 500 vendors**. The large number of devices means lots of opportunity to automate my apartment. An ESP32 can also be programmed to support Zigbee.

A Raspberry Pi will be used as the central host to coordinate across devices within the network.

Example Topology

Coordinator:

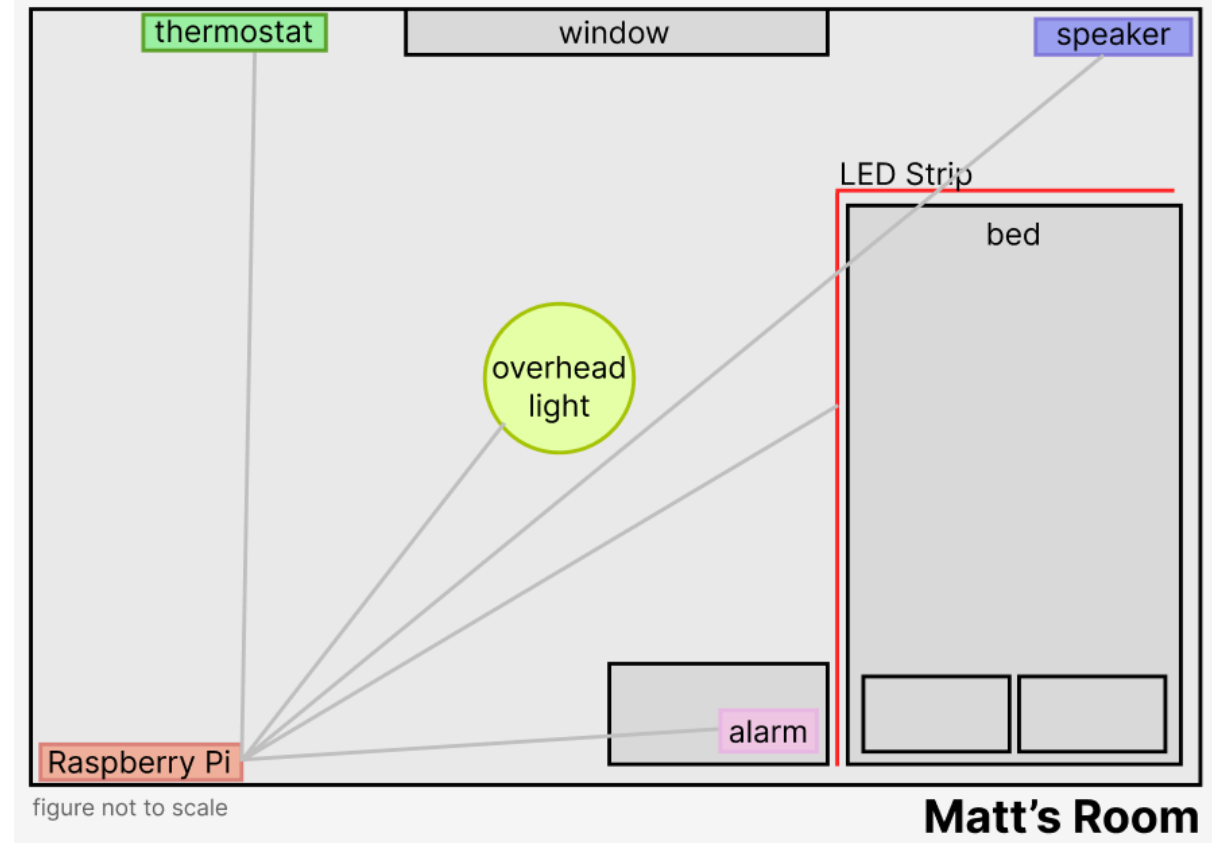
- Raspberry Pi

End Devices:

- Overhead Lightbulb
- LED Strip
- Thermostat
- Alarm Clock

Non-Zigbee Device:

- Speaker (this will require a BT signal)



This creates a **Star Topology** which will work well within the small space of my room (<100m)

Example Implementation (Morning Routine)

The first automation I would create a **simple morning automation** to help to not snooze in the morning. (something I struggle with 😊). This uses automated lighting and temperature setting to help me start the day!

Functionality:

1. 30 minutes before wakeup:
 - raise the temperature to 72 deg (from 63)
2. When wakeup occurs:
 - Raise alarm
 - Turn on LED strip to red
 - Play "Good Morning" by Kanye West
3. After song end:
 - Turn off LED strip
 - Turn on Lightbulbs
 - Set temperature to 68 deg

Components:

- Zigbee Lightbulb
- Zigbee LED Strip
- Zigbee Alarm Clock
- Bluetooth Speaker

[Zigbee Device List](#)

Morning Automation



Radiator valve with thermostat

Alecto

SMART-HEAT10 - child_lock, switch, battery, climate

6:30am:

- set thermostat to 72 degrees from 63 degrees to raise body temperature



Temperature and humidity sensor with clock

Tevo

TZE200_13xd744 - temperature, humidity, battery, temperature, unit



Hue White and Color Flux Strip Light 10m

new Philips

929004810382 - light, power_on, behavior



7:00 am (WAKEUP):

- raise alarm for 5 seconds before playing “Good Morning”
- turn on red LEDs

...song is playing...



Hue White and Color Flux Strip Light 10m

new Philips

929004810382 - light, power_on, behavior



ENK LEXMAN E27 LED white

ADEO

1A-CD2FB2AA057N6-M2N-01 - light, power_on, behavior



Radiator valve with thermostat

Alecto

SMART-HEAT10 - child_lock, switch, battery, climate

7:03 am:

- turn off red LEDs
- turn on overhead lights
- set thermostat to 68 degrees

Example Implementation (Nighttime Routine)

The second automation I would create a **simple nighttime automation** to help me fall asleep quicker.

This is very similar to the morning routine, using automated lighting and temperature setting to help me start the day!

Functionality:

1. 30 minutes before sleep:
 - drop the temperature to 63 deg (from 68)
 - Turn on LED strip to red
2. When bedtime occurs:
 - Turn on "White Noise"
 - Turn off LED strip

Components:

- Zigbee Lightbulb
- Zigbee LED Strip
- Zigbee Alarm Clock
- Bluetooth Speaker

[Zigbee Device List](#)

Nighttime Automation



9:30 pm:

- set thermostat to 63 degrees from 68 degrees to lower body temperature
- turn on red LEDs

10:00 pm (WAKEUP):

- turn off red LEDs
- turn on white noise

Conclusion

Project Retrospective and Thanks

Project Retrospective

What was good:

This project was reading heavy as I incorporated many sources to gain enough information to compare each of the three protocols in detail. In this regard, it was a success as I now understand key differences between the three protocols. Additionally, I answered my question of which protocol I will personally use for future smart home development – Zigbee!

What could have been improved upon:

However, this project lacked a tangible result. During my project proposal, your feedback was to look into setting up a testbed/experiment. I did not take this route due to my semester workload and part-time work obligations. To compensate for this, I did design a sample automation that is feasible using Zigbee.

Thank you, Professor Zhang!

Topics in Computer Science II has given me perspective on how the digital and physical worlds interacts in ways which no other class has given me. The underlying topics of computer networking, scheduling policies, and data transmission has given me a deeper understanding of how RTCPs works - and the intuition to have an educated guess on a glance.

Also, thank you very much for your dedication to us students! I am super appreciative of the thoughtful and actionable feedback you gave during each of our presentations. This is advice I will take with me into industry.

Enjoy your winter break!

Sincerely,
Matt Krueger

IOWA

Primary Sources

Thread:

- [1] [Thread Fundamentals](#)
- [2] [Thread Mesh Network Performance](#)

Zigbee:

- [3] [Lecture 8 Zigbee and WirelessHART \(Topics in Computer Science II\)](#)
- [4] [Zigbee Mesh Network Performance](#)
- [5] [Analysis on the Security Architecture of ZigBee Based on IEEE 802.15.4](#)
- [6] [Getting Started with Zigbee for STM32WB and STM32WBA Series](#)

Z-Wave:

- [7] [Introduction of Z-Wave](#)
- [8] [Smart Homes Automation using Z-wave Protocol](#)
- [9] [Z-Wave Long Range](#)

Secondary Sources

Wikipedia:

- [10] [Zigbee](#)
- [11] [Z-Wave](#)
- [12] [Thread](#)

Device Counts:

- [13] [Zigbee](#)
- [14] [Z-Wave](#)
- [15] [Thread](#)

Memberships:

- [16] [Thread Group](#)
- [17] [Connectivity Standards Alliance](#)
- [18] [Z-Wave Alliance](#)

Assets

Images:

- [Thread logo](#)
- [Zigbee logo](#)
- [Z-Wave logo](#)
- [Kanye West "Good morning"](#)
- [White noise](#)

University of Iowa:

- [PowerPoint Template](#)